THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 10: **Light: Shadows and Reflections** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A ray of light falls on a plane mirror along the normal. The ray: 1

(a) passes through the mirror **(b)** travels along the mirror surface
(c) returns along the same path **(d)** is absorbed

2. During a solar eclipse, the Moon's shadow falls on the Earth. The region with total darkness is the: 1

(a) penumbra **(b)** umbra **(c)** halo **(d)** corona

3. Why does the image in a mirror appear to be as far behind as the object is in front? 1

(a) light bends
(b) virtual image position follows the law of reflection
(c) mirror is smaller
(d) mirror is thick

4. If the angle of incidence of a ray on a plane mirror is increased by 15°, the angle of reflection: 1

(a) increases by 15° **(b)** stays the same **(c)** decreases by 15° **(d)** doubles

5. During a lunar eclipse, the Moon passes into Earth's: 1

(a) orbit only **(b)** shadow **(c)** atmosphere only **(d)** light

6. A red rose in white light is viewed in a plane mirror. The image of the rose appears: 1

(a) green **(b)** black **(c)** red **(d)** white

7. The image formed by a plane mirror of a person 1.6 m tall is: 1

(a) 0.8 m tall **(b)** 0.4 m tall **(c)** 3.2 m tall **(d)** 1.6 m tall

8. A student sees a shadow of her hand on a screen 2 m from a lamp. If she moves the hand to the midpoint between lamp and screen, the shadow size relative to the hand is: 1

(a) same **(b)** half **(c)** four times **(d)** double

9. **Assertion (A):** A shadow is formed when an opaque object comes in the path of light. 1
Reason (R): Light travels in straight lines.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** The Moon is a non-luminous object. 1
Reason (R): The Sun is a luminous object.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A ray is incident at 90° to a plane mirror. State the angle of incidence and the path of the reflected ray. 2

12. Why is the shadow of a tall pole longer in the evening than at noon? 2

13. Explain why the Moon looks bright at night but is not a luminous body. 2

14. A plane mirror forms an image of a 5 cm tall object. What are the image height and nature? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A ray of light strikes a mirror with an incident angle of 55°. Find the angle between the incident ray and the reflected ray, and the angle between the mirror and the reflected ray. 3

16. Describe how to use a stick and its shadow to find the time of noon. 3

17. Why does a wet black road at night show the lights of vehicles? 3

OR

A girl is 4 m from a mirror and walks 1 m towards it. By how much does the distance between her and her image decrease? 3

18. Why are reflecting surfaces on roads and bicycles used? 3

OR

Explain why a mirror placed in a dark room cannot be seen. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Arun makes hand shadows of a bird on the wall of his room using a torch. When he moves his hand closer to the torch, the shadow becomes bigger. He also tries a sheet of clear glass and a sheet of butter paper in front of the torch. 5
(i) Write the three things needed to form a shadow.
(ii) What happens to the size of the shadow when the object is moved nearer to the torch?
(iii) What type of material is butter paper?
(iv) Will the clear glass sheet form a dark shadow? Give a reason.

20. Design an experiment to find the relation between the object distance and the image distance in a plane mirror, with a table and conclusion. 5

OR

Explain how a plane mirror in a barber's shop shows the back of a customer's head, using a second mirror. 5

END OF QUESTION PAPER — Total: 40 Marks